

Homework 3 Write-up

Part I: EC2 and PySpark Setup:

I created an EC2 instance (id: i-0870e65ecd7e6d126) running Amazon Linux, in the us-east-1 region.

I also installed the following dependencies:

- Java 17 (Amazon Corretto)
- PySpark
- pandas, pyarrow
- matplotlib

The NYC TLC trip data was copied directly from the given S3 bucket in the homework document to my EC2 instance using AWS CLI.

Part II: Dataset Used:

The dataset used for this homework is the NYC TLC trip records from yellow and green taxis for January 2026, as per the assignment example.

Part III: Data Cleaning Assumptions:

For data cleaning, the following assumptions were applied to both the yellow and green taxi datasets:

- Removing all rows with missing pick-up or drop-off timestamps.
- Removing all trips with negative or zero distance values.
- Removing any trips with negative fare amounts.
- Removing any trips with negative total amounts.
- Removing trips if the drop-off time preceded the pick-up time.
- Removing any trips that were longer than 24 hours.

After cleaning, the yellow taxi dataset had 3,518,128 rows (with 206,761 rows removed), and the green taxi dataset had 38,909 rows (with 1,363 removed).

Furthermore, as part of standardizing the datasets before combining, the datetime columns were renamed and only shared columns were included before combining the two datasets.

Part IV: Analytics Questions:

Question 1: Which taxi type had more trips?

Yellow taxis had a significantly higher number of trips than the green taxis.

```
Q1: Which taxi type had more trips?  
Yellow: 3,518,128, Green: 38,909
```

Question 2: What was the average fare by taxi type?

The average fare for yellow taxis was about \$21.08, while the average fare for green taxis was \$16.00.

```
Q2: Average fare by taxi type  
Yellow: $21.08, Green: $16.0
```

Question 3: What was the average trip distance by taxi type?

The green taxis had almost double the average trip distance of yellow taxis, which makes sense because green taxis operate in the outer boroughs of NYC where the trip distances tend to be longer.

```
Q3: Average trip distance by taxi type  
Yellow: 6.76 miles, Green: 12.99 miles
```

Question 4: What hour of day had the most pickups?

6:00 PM had the most pickups, due to expected post-working hours rush.

```
Q4: Hour of day with most pickups  
Hour 18 (6 PM) with 232,570 pickups
```

Question 5: What percentage of the trips were under 2 miles?

51.95% of all trips were under two miles.

```
Q5: Percentage of trips under 2 miles  
51.95%
```

Question 8: Fare Amount Prediction Using Linear Regression

A linear regression model was trained to predict the fare amount. All non-fare predictor columns were taken into account including: trip_distance, passenger_count, PULocationID, DOLocationID, RatecodeID, congestion_surcharge, cbd_congestion_fee, improvement_surcharge and mta_tax. As advised in the assignment, the data was split 80/20 for training and testing.

The following were the RMSE values for both training and testing:

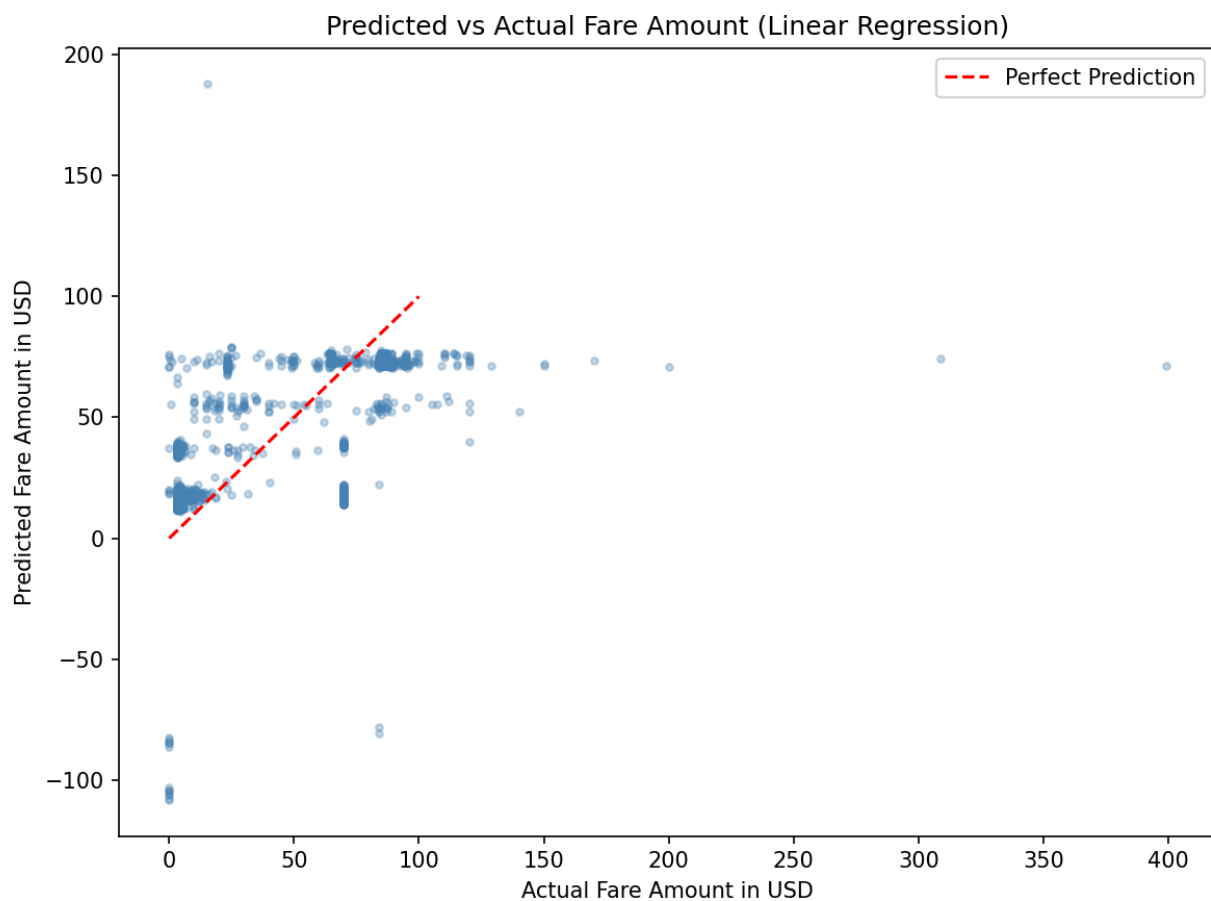
```
Training RMSE: 15.865214196147333  
Testing RMSE: 15.988019620833336
```

Since the training and testing RMSE values are close, that means our model is not overfitting. However, the high RMSE of almost \$16 probably indicates that linear regression is too simple and cannot fully pick up fare patterns.

In addition to RMSE values, the predictor coefficients were also calculated from the trained model as shown in the screenshot below. The largest coefficients were the improvement_surcharge and the mta_tax, even though both are fixed fee columns. On the contrary, trip_distance had a much weaker effect.

```
Predictor Coefficients:  
trip_distance: 0.011293819591222446  
passenger_count: 1.1290985087825993  
PULocationID: -0.016299645498689085  
DOLocationID: -0.003467251275166554  
RatecodeID: 1.534232130108943  
congestion_surcharge: -8.653490161099876  
cbd_congestion_fee: 4.330633428113276  
improvement_surcharge: 150.80703039337496  
mta_tax: -61.138713168520354
```

Finally, the plot below shows the predicted fare vs the actual fare amount and it clearly shows the aforementioned, especially in that linear regression struggles in this use case. A tree based model would have most likely performed better.



Part V: S3 Output Paths:

- ***Trip count by taxi type:*** s3://gomaa-hw3/results/trips_by_type/
- ***Average fare by taxi type:*** s3://gomaa-hw3/results/avg_fare_by_type/
- ***Pickups by hour:*** s3://gomaa-hw3/results/pickups_by_hour/
- ***Fare prediction plot:*** s3://gomaa-hw3/fare_prediction.png